

Breech Delivery + NRP



Section I: Scenario Demographics

Scenario Title:	Breech Delivery + NRP		
Date of Development:	29/02/2016 (DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 – 2)	☒ Seniors (PGY ≥ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Donika Orlich (NRP from NRP case by Kyla Caners)
Affiliations/Institution(s):	McMaster University
Contact E-mail (optional):	Donika.orlich@medportal.ca

Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners to a delivery with breech presentation and a neonate requiring resuscitation.
CRM Objectives:	1) Demonstrates leadership by staying calm during crisis, making prompt decisions and maintaining a global perspective 2) Demonstrates communication skills by listening to team input and updating mother on status of infant 3) Demonstrates resource utilization by calling for help appropriately, prioritizing tasks and triaging patients as required
Medical Objectives:	1) Anticipates emergent delivery and prepares necessary equipment 2) Demonstrates an approach to fetal delivery & management of breech presentation 3) Appropriately manages the flat neonate using the NRP guidelines for O2 use, PPV, intubation, chest compressions and medications

Case Summary: Brief Summary of Case Progression and Major Events

A 19-year-old female presents with EMS in active labour. She denies any history of pregnancy and has had no prenatal care. On examination, infant will be in breech position. The learner must deliver the infant from breech presentation. Following this, the neonate will present lifeless, and require resuscitation.

References

Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.



Breech Delivery + NRP



Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☐ Adult Computerized Mannequin	Realism:	☒ Conceptual
	☒ Neonatal Computerized Mannequin		☒ Physical
	☒ Standardized Patient PLUS OB task trainer (hybrid)		☒ Emotional/Experiential
	☐ Other		☐ Other:
		<i>Select most important dimension(s)</i>	☐ N/A
Con-federates	Brief Description of Role		
SP	Active labor. Heavy breathing + pain during contractions. Later very concerned about baby, crying, must be consoled		
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Adult + Neonatal Bag Valve Mask	☒ Ultrasound machine	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☒ Blood Products	☒ Adult + Neonatal ET Tubes	☒ Other: Umbilical vein catheter	
☒ Interosseous Set-up	☐ LMA	☐ Other: Scissors, umbilical cord clamps, blankets, neonatal warmer	
D. Moulage			
Blood/Fluid covering neonate initially. Task trainer set up with baby in breech position.			
E. Approximate Timing			
Set-Up:	10 min	Scenario:	20 min
		Debriefing:	40 min

Breech Delivery + NRP



Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case			
EMS Patch: "We have a 19 F complaining of severe abdominal pain onset 1 hour ago. She denies being pregnant, but looks almost full term to us. Contractions seem to be about 1 minute apart. Patient's Vitals as follows: HR 120, BP 140/85, RR 20, O2 100% on RA. ETA 2 minutes."			
B. Patient Profile and History			
Patient Name: Naveah Scott		Age: 19	Weight: 90kg
Gender: ☐ M ☒ F		Code Status: Full	
Chief Complaint: Abdominal Pain			
History of Presenting Illness: Started with cramping about 1 hour ago. Thought it might be her period, but this is the worst pain of her life. Seems to come in waves every few minutes.			
Past Medical History:	Healthy	Medications:	None
	No prior pregnancies.		
	No prenatal care ("I can't be pregnant")		
Allergies: None			
Social History: Lives in a shelter after leaving home at 16. No smoking. Occasional EtOH. No drugs.			
Family History: None			
Review of Systems:	CNS:	Visible pain/discomfort.	
	HEENT:	Nil	
	CVS:	Nil	
	RESP:	Nil	
	GI:	Nil	
	GU:	Severe abdo cramping q1min. Fluid gush with EMS. No bleeding. Denies every having felt a baby move.	
	MSK:	Nil	INT: Nil
C. Baseline Simulator State and Physical Exam			
☐ No Monitor Display		☒ Monitor On, no data displayed	
☐ Monitor on Standard Display			
HR: 120/min	BP: 125/80	RR: 22/min	O ₂ SAT: 100%
Rhythm: Sinus Tach	T: 36.5°C	Glucose: 6 mmol/L	GCS: 15 (E V M)
General Status: Visible pain/discomfort. Heavy breathing.			
CNS:	Alert and Oriented		
HEENT:	Normal.		
CVS:	Normal heart sounds. No edema.		
RESP:	GAEB. No crackles/wheeze.		
ABDO:	Leopold's: Fundus under ribs, back to mothers left, head UP. FHR: 150 bmp, little variability, late decelerations		
GU:	Sterile Cervical Assessment: feel fetal buttocks		

Breech Delivery + NRP



Section VI: Scenario Progression

Scenario States, Modifiers and Triggers- PATIENT #1 (Mother)			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. EMS Patch		<u>Learner Actions</u> ☐ Prepare for delivery (scissors, clamps, oxytocin, foley, analgesic, suture/needle driver) ☐ U/S machine in room ☐ Prepare for NRP (warmer, blankets, O2, braselow, ETT/suction, infant BMV) delegate team member to baby's management ☐ Early consult to Obstetrics + NICU + Anesthesia ☐ Pull nursing/RT staff to increase available team members	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> <u>Triggers</u> <i>For progression to next state</i> - 2 minutes → 2. Patient Arrival
2. Patient Arrival Rhythm: Sinus tach HR: 122/min BP: 138/80 RR: 22/min O ₂ SAT: 100% FHR: 140/min	++ Discomfort, heavy breathing, crying with contractions.	<u>Learner Actions</u> ☐ IV, O2, monitors ☐ Px: Leopold's, FHR monitor, sterile cervical examination ☐ US (confirm presenting part) ☐ Labs (CBC, G&S, INR) ☐ Focused Hx	<u>Modifiers</u> - No attempt at cervical exam by 2mins → patient screams "I feel something down there!!" <u>Triggers</u> - 3 mins or all learner actions complete → 3. Delivery
3. Delivery Rhythm: Sinus tach HR: 130/min BP: 140/90 RR: 28/min O ₂ SAT: 100% FHR: 100	++ Screaming in pain	<u>Learner Actions</u> ☐ Provide IV analgesia ☐ Mediolateral episiotomy ☐ Check for cord prolapse ☐ Flex knees & deliver legs/hips ☐ After umbilicus delivers, pull out 10-15cm of cord ☐ Deliver arms (keep face & abdomen away from symphysis) ☐ Mauriceau maneuver to deliver the head (check nuchal cord first) ☐ Oxytocin 10U IM ☐ Clamp and cut cord ☐ Infant to NRP provider	<u>Modifiers</u> - Nuchal cord not checked before delivering head → FHR to 80 bpm <u>Triggers</u> - Infant delivered → 4. Post-delivery AND Begin PATIENT #2 Case

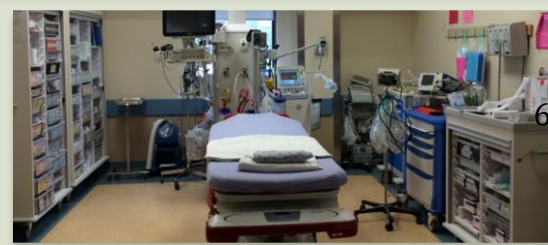


Breech Delivery + NRP



4. Post-Delivery HR: 105 BP: 120/80 RR: 12 O2: 100%	Mother crying and asking about her baby. Otherwise comfortable.	<u>Learner Actions</u> - ☐ Stay with mother for placental delivery (NRP delegated to another provider) - ☐ Gentle traction and SP pressure until placenta delivered - ☐ Inspect vessels and placenta integrity - ☐ Trans-abdominal uterine massage - ☐ Update mother +/- SW consult	<u>Modifiers</u> - Mother not updated → begins to wail and cry “what is happening to my baby” and tries to get out of the bed END CASE WHEN PATIENT #2 CASE ENDS
--	---	---	---

Breech Delivery + NRP



Scenario States, Modifiers and Triggers- PATIENT #2 (Neonate)			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: sinus HR: 40/min BP: ?/? RR: 20/min O ₂ SAT: 58% T: 36.5°C	Flat. Not crying. Minimal respiratory effort. No meconium	<u>Learner Actions</u> - ☐ Dry & warm - ☐ Position airway - ☐ Consider suctioning (mouth then nose) - ☐ Stimulate - ☐ Check HR → start PPV with 21% O ₂ (rate 40-60/min) - ☐ Initiate SpO ₂ monitoring - ☐ Check cap sugar: 5.2	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - PPV → O ₂ 65%, HR 45 <u>Triggers</u> <i>For progression to next state</i> - 1 min → 2. 60 Second Mark
2. 60 Second Mark HR:45 RR: bagging rate O ₂ : 65% *same as above if no BMV started	Unchanged.	<u>Learner Actions</u> - ☐ Re-check HR at 60 sec - ☐ Start CPR at 90 sec - ☐ Change to 100% O ₂ - ☐ Maintain 3:1 ratio for CPR (120 events/min) - ☐ Check HR/RR q1min - ☐ Epi 0.01mg/kg IV after 1 st HR check if still <60 after CPR - ☐ Prepare for intubation	<u>Modifiers</u> - 2 min → O ₂ SAT 65% - 3 min → O ₂ SAT 70% <u>Triggers</u> - 4 min → 3. HR >60
3. HR > 60 HR → 80 RR → 40 O ₂ SAT → 75% BP → 55/30	Appears less grey.	<u>Learner Actions</u> - ☐ Intubate - ☐ Obtain iv/io access - ☐ Prepare for UVC placement - ☐ IVF bolus (20ml/kg =50ml) - ☐ Check cap sugar (if not yet) - ☐ Send blood work, including blood cultures	<u>Modifiers</u> - 5 min → O ₂ SAT 80% <u>Triggers</u> - 7 min → 4. HR >100 (set to slowly return to normal)
4. HR >100 O ₂ SAT → 83% HR → 120	Appears pink. Moving limbs now.	<u>Learner Actions</u> - ☐ Call NICU - ☐ Insert UVC - ☐ AXR to confirm UVC placement, CXR for ETT - ☐ Finish tasks from above	<u>Triggers</u> - 10 minutes or once intubated/UVC placed → END PATIENT #2 case

Breech Delivery + NRP



Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results- PATIENT #1	
None Given	

Images (ECGs, CXRs, etc.)	
None available.	

Ultrasound Video Files (if applicable)	
Needed- US breech (RN to tell learners that they see buttocks down and head up)	



Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
Educational Goal:	To expose learners to a delivery with breech presentation and a neonate requiring resuscitation.
CRM Objectives:	1) Demonstrates leadership by staying calm during crisis, making prompt decisions and maintaining a global perspective 2) Demonstrates communication skills by listening to team input and updating mother on status of infant 3) Demonstrates resource utilization by calling for help appropriately, prioritizing tasks and triaging patients as required
Medical Objectives:	1) Anticipates emergent delivery and prepares necessary equipment 2) Demonstrates an approach to fetal delivery & management of breech presentation 3) Appropriately manages the flat neonate using the NRP guidelines for O2 use, PPV, intubation, chest compressions and medications
Sample Questions for Debriefing	
1) What was your approach to managing two simultaneous resuscitations? How did you delegate tasks? Did you use any safety parameters with those you delegated to? 2) What communication strategies did you use to calm the mother? 3) What are the steps involved in delivering a breech baby? 4) When would you consider narcan in a neonate?	
Key Moments	
1) Recognizes breech presentation and employs a tailored approach to delivery of the infant. 2) Recognizes need to task switch to infant requiring resuscitation and effectively hands over maternal care to junior staff. 3) Initiates NRP using current guidelines for O2, PPV, intubation, chest compressions and epinephrine use.	